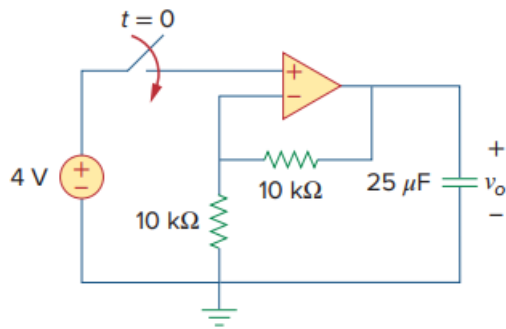


Problem

Obtain v_o for $t > 0$ in the circuit of Fig. 7.133.

Fig. 7.133:



Step-by-step solution

Step 1 of 9

Consider the circuit as shown in Figure 1.

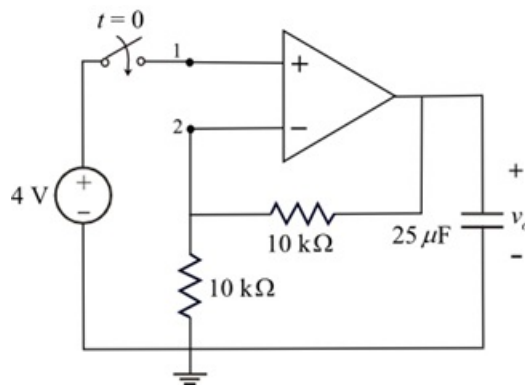


Figure 1

Step 2 of 9

Initially the switch is opened for a long time. So, consider the circuit for $t < 0$ as shown in Figure 2.

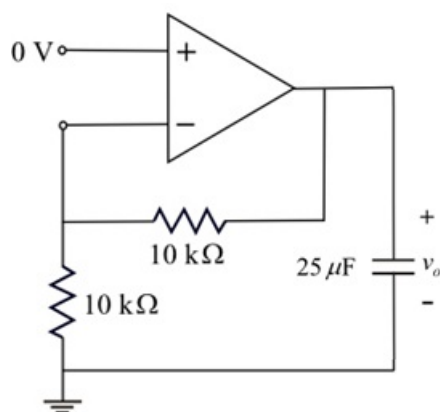


Figure 2

Step 3 of 9

The general equation for the voltage across the capacitor is,

$$v_o(t) = v_o(\infty) + [v_o(0) - v_o(\infty)]e^{-\frac{t}{\tau}} \dots\dots (1)$$

The Figure 2 is a source free network. Therefore,

$$v_o(0^-) = 0 \text{ V}$$

The voltage across the capacitor cannot change instantaneously. Therefore,

$$v_o(0^-) = v_o(0) = 0 \text{ V}$$

Step 4 of 9

At $t = 0$, the switch is closed and the $25 \mu\text{F}$ capacitor acts as a short circuit.

Hence, for $t > 0$, consider the circuit as shown in Figure 3.

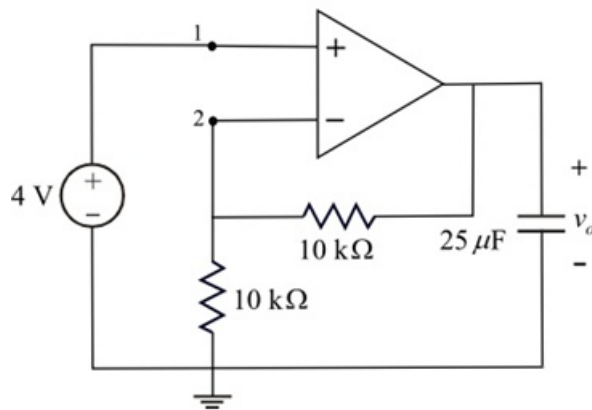


Figure 3

Step 5 of 9

At $t = \infty$, the circuit reaches steady state and the capacitor becomes open circuited as shown in Figure 4.

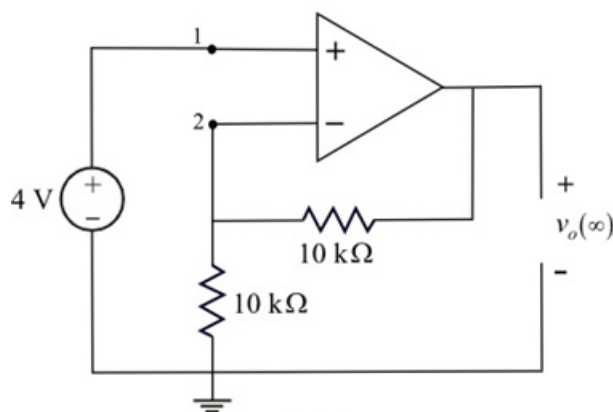


Figure 4

Step 6 of 9

Find the voltage $v_o(\infty)$.

From the Figure 4, the voltage at the non-inverting terminal of the op-amp is,

$$v_1 = 4 \text{ V}$$

Assume that the op-amp is ideal. The feedback given is the negative feedback. Hence, by virtual ground concept, the voltage at the inverting and non-inverting terminals is same. That is,

$$v_1 = v_2 = 4 \text{ V}$$

Apply Kirchhoff's current law equation at node v_2 .

$$\frac{v_2 - v_o(\infty)}{10 \text{ k}} + \frac{v_2}{10 \text{ k}} = 0$$

$$\frac{v_2 - v_o(\infty) + v_2}{10 \text{ k}} = 0$$

$$2v_2 - v_o(\infty) = 0$$

$$v_o(\infty) = 2v_2$$

Substitute 4 V for v_2 in the equation.

$$\begin{aligned} v_o(\infty) &= 2(4) \\ &= 8 \text{ V} \end{aligned}$$

Step 7 of 9

Find the time constant τ .

The time constant of the circuit is,

$$\tau = R_{\text{Th}} C$$

Here, R_{Th} is the Thevenin's equivalent resistance.

To find R_{Th} , consider the circuit as shown in Figure 5.

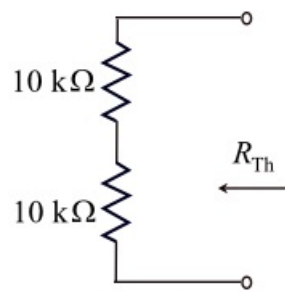


Figure 5

Step 8 of 9

From the Figure 5, the Thevenin's resistance is,

$$\begin{aligned} R_{\text{Th}} &= 10 \text{ k}\Omega + 10 \text{ k}\Omega \\ &= 20 \text{ k}\Omega \end{aligned}$$

The time constant of the circuit is,

$$\begin{aligned} \tau &= R_{\text{Th}} C \\ &= (20 \text{ k}\Omega)(25 \times 10^{-6}) \\ &= 500 \times 10^{-3} \text{ s} \\ &= 0.5 \text{ s} \end{aligned}$$

Step 9 of 9

The expression for v_o for $t > 0$ is,

$$v_o(t) = v_o(\infty) + [v_o(0) - v_o(\infty)]e^{-\frac{t}{\tau}}$$

Substitute 0 V for $v_o(0)$, 8 V for $v_o(\infty)$, 0.5 s for τ .

$$\begin{aligned}v_o(t) &= 8 + [0 - 8]e^{-\frac{t}{0.5}} \\&= 8 - 8e^{-2t} \\&= 8(1 - e^{-2t}) \text{ V}\end{aligned}$$

Therefore, the voltage v_o for $t > 0$ is $\boxed{8(1 - e^{-2t}) \text{ V}}$.